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Web Application for Managing Borehole Data Based on SpringBoot Framework

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Web Application for Managing Borehole Data Based on SpringBoot Framework

Abstract. A borehole is a generalized term for any narrow shaft drilled in the ground, either vertically or horizontally. They are used as part of a geotechnical investigation. Gathered data enable the preparation of statistics describing the composition of the soil, and distinct geological layers having varying properties. In order to make the analysis of the borehole data more concise and intuitive, a web application to visualize the borehole data and summarize the boreholes was designed.

The system is based on Browser/Server architecture. It is based on Java, HTML, CSS, JavaScript. The back-end is created using the development framework SpringBoot. Front-end was written using development framework VUE and VUE's front-end component library Element-UI. The data is stored in MySQL database.

This system performs the functions of reading borehole data from CSV files and displays the data both in table format as well as map visualization. It allows the users to create new borehole data, modify existing borehole data, and filter the data by single or multiple conditions. Users can export the data to a CSV file or export the map containing selected borehole locations as image.

The tests confirm that developed system runs smoothly and improves the efficiency of borehole data management and reduces the difficulty of analysis.

Keywords: Web Application, SpringBoot, VUE, MyBatis-Plus, OpenLayers, Borehole data management

Aplikacja internetowa do zarządzania danymi o odwiertach na bazie szkieletu aplikacyjnego SpringBoot

Streszczenie. Odwiert to uogólnione określenie każdego wąskiego szybu wywierconego w ziemi, pionowo lub poziomo. Wykorzystywane są w ramach badań geotechnicznych. Zgromadzone dane umożliwiają przygotowanie statystyk opisujących skład gruntu oraz wyodrębnione warstwy geologiczne o zróżnicowanych właściwościach. Aby analiza danych z odwiertów była bardziej zwięzła i intuicyjna, zaprojektowano aplikację internetową do wizualizacji danych z odwiertów i ich przeglądu.

System oparty jest na architekturze Przeglądarka/Serwer. Opiera się na Java, HTML, CSS, JavaScript. Back-end jest tworzony przy użyciu frameworka programistycznego SpringBoot. Front-end został napisany przy użyciu frameworka programistycznego VUE i biblioteki komponentów front-endowych VUE Element-UI. Dane są przechowywane w bazie danych MySQL.

System ten realizuje funkcje odczytu danych z odwiertów z plików CSV i wyświetla dane zarówno w formie tabelarycznej, jak i wizualizacji położenia na mapie. Umożliwia użytkownikom tworzenie nowych danych odwiertów, modyfikację istniejących danych oraz filtrowanie danych według jednego lub wielu warunków. Użytkownicy mogą eksportować dane do pliku CSV lub eksportować mapę zawierającą wybrane lokalizacje otworów wiertniczych jako obraz.

Testy potwierdzają, że opracowany system działa płynnie i poprawia efektywność zarządzania danymi wiertniczymi oraz zmniejsza trudność analizy.

Słowa kluczowe: Web Application, SpringBoot, VUE,MyBatis-Plus, OpenLayers, zarządzanie danymi odwiertów

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1. Introduction

1.1. Background

A borehole is a generalized term for any narrow shaft drilled in the ground, either vertically or horizontally. A borehole may be drilled for a variety of purposes, such as the extraction of water and other liquids or gases, as part of a geotechnical investigation or environmental site assessment, for the exploration of minerals, or as a pilot hole for installing piers or underground utilities. The geological boreholes from the Engineering-Geological Atlas of Warsaw Database have more than 27, 000 records [1]. The variety and complexity of borehole data make it more difficult to read, but tabularization and map visualization can enhance this. This thesis is devoted to designing and implementing a web application system for data management based on the SpringBoot+MyBaties-plus framework, which improves the convenience and standardization of borehole data management.

1.2. Related Work

The development of this system is based on the concept of object-oriented, learning knowledge and integrating related technologies, this development makes use of the idea of separating of front-end and back-end of the architecture so that the front-end and back-end are more focused while developing and enhancing the development efficiency [2]. For the back-end, Java language and IDEA compiler integrated development environment were used, with the help of SpringBoot and MyBaties-plus frameworks, which makes the development process more convenient. MySQL is used for storing data inside the database. For the front-end, HTML+CSS+VUE were used to implement the page view. For the map visualization part, which is also the focus of this system, the OpenLayers opensource JavaScript library was used to embed the map into the web page via the relevant API. The project build tool uses Maven to manage Jar packages and project builds. After completing the development of the system, the testing was carried out by way of software engineering testing techniques, to verify every feature, and make refinements and improvements.

1.3. Structure

The structure details of this thesis are shown below:

Chapter 1 Introduction: it contains a background introduction to the creation of this thesis and the motivation for its creation, explaining the importance of the web application. An overview describes the related work in this thesis and briefly describes the overall architecture as well as frameworks and technologies used in the project.

Chapter 2 Technologies and tools: analysis of relevant theories and key technologies as well as develop tools and environment. This includes the front-end related VUE framework, the back-end related SpringBoot framework, and the database related MyBaties-plus

6. Conclusion

The design and implementation of the borehole data management system are the main topics discussed in this thesis. Following an introduction to the project's background and the motivation behind its conception, a detailed description of the system's technologies is provided. The system's specific requirements are next analyzed and presented from a standpoint of usability. The system design, database design, and system implementation are then accomplished and described in detail. Finally, the system was tested to ensure that it complied with all the requirements, and during the test, it had further improvements.

In terms of system capabilities, we can upload and read CSV files, manage borehole data on the map, and manage borehole data on the table, for example: add new data, update data, etc., and export data. It makes observing and analyzing borehole data much easier. In terms of project architecture, we combine SpringBoot and MyBatis-plus back-end frameworks, VUE and Element-UI front-end frameworks, using the front-end and back-end separation of development, with other major development plugins, improved development efficiency, enhanced system scalability, changed the previous overall packaging method, and achieved better maintenance.

Up to this point, the system's expected features have essentially been implemented, and the results of the corresponding data analysis are broadly consistent with expectations. However, there are still rooms for improvement. In terms of system functionality, user management features can be added to grant users different permissions to do different tasks. The ability for multiple people to evaluate and analyze the same group of data when the data set is too big is essential, since it can significantly increase efficiency. The system will also become more robust with features like uploading and overlaying multiple files, customization of borehole data point markers, and three-dimensionalization of maps. There are still more advanced and effective technologies in system development that I need to learn in order to advance to a higher level in the area of web application development.

References

[1] S. Jankowski, A. Hrechka, Z. Szymański, and G. Ryżyński, "Modeling engineering-geological layers by k-nn and neural networks", in *International Conference on Neural Networks and Artificial Intelligence*, Springer, 2014, pp. 147–158.

[2] Y. Gong, F. Gu, K. Chen, and F. Wang, "The architecture of micro-services and the separation of frond-end and back-end applied in a campus information system", in *2020 IEEE International Conference on Advances in Electrical Engineering and Computer Applications(AEECA)*, 2020, pp. 321–324. DOI: 10.1109/AEECA49918.2020.9213662.

[3] Y. Liu, "Design and implementation of network structure visualization system based on b/s architecture", in *Proceedings of the 10th International Conference on Communications and Broadband Networking*, ser. ICCBN '22, Shanghai, China: Association for Computing Machinery, 2022, pp. 14–19, ISBN: 9781450387439. DOI: 10.1145/3538806.3538813. [Online]. Available: https://doi.org/10.1145/3538806.3538813.

[4] K. Yang, "The network security analysis system design based on b/s structure", in *Proceedings of the 2018 International Conference on Internet and E-Business*, ser. ICIEB '18, Singapore, Singapore: Association for Computing Machinery, 2018, pp. 170–173, ISBN: 9781450363754. DOI: 10.1145/3230348.3230451. [Online]. Available: https://doi.org/10.1145/3230348.3230451.

[5] N. Li and B. Zhang, "The research on single page application front-end development based on vue", *Journal of Physics: Conference Series*, vol. 1883, no. 1, p. 012 030, Apr. 2021. DOI: 10.1088/1742-6596/1883/1/012030. [Online]. Available: https://doi.org/10.1088/1742-6596/1883/1/012030.

[6] E. Wohlgethan, "Supportingweb development decisions by comparing three major javascript frameworks: Angular, react and vue. js", Ph.D. dissertation, Hochschule für Angewandte Wissenschaften Hamburg, 2018.

[7] J. Arthur and S. Azadegan, "Spring framework for rapid open source j2ee web application development: A case study", in *Sixth International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing and First ACIS International Workshop on Self-Assembling Wireless Network*, 2005, pp. 90–95. DOI: 10.1109/SNPD-SAWN.2005.74.

[8] X. Luan, "Implementation and analysis of software development in spring boot", Ph.D. dissertation, California State Polytechnic University, Pomona, 2021.

[9] M. Błaszczyk, M. Pucek, and P. Kopniak, "Comparison of lightweight frameworks for java by analyzing proprietary web applications", *Journal of Computer Sciences Institute*, vol. 19, pp. 159–164, 2021.

[10] L. Chung and J. C. S. d. Prado Leite, "On non-functional requirements in software engineering", in *Conceptual modeling: Foundations and applications*, Springer, 2009, pp. 363–379.

[11] J. J. Garrett *et al.*, "Ajax: A new approach to web applications", 2005.
[12] *A high-performance, feature-packed library for all your mapping needs.* [Online]. Available: `https://openlayers.org/`.
[13] *Html2canvas.* [Online]. Available: `https://html2canvas.hertzen.com/`.
[14] M. Esnaashari and A. H. Damia, "Automation of software test data generation using genetic algorithm and reinforcement learning", *Expert Systems with Applications*, vol. 183, p. 115 446, 2021.
[15] R. A. DeMillo, "Software testing", in *Encyclopedia of Computer Science*. GBR: John Wiley and Sons Ltd., 2003, pp. 1645–1649, ISBN: 0470864125.
[16] I. Jovanović, "Software testing methods and techniques", *The IPSI BgD Transactions on Internet Research*, vol. 30, 2006.

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